

Logical Operators Quick Reference				
$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$
$\forall$	$\exists$	$\in$	$\notin$	$\equiv$

1A. $(\forall x(C(x) \rightarrow M(x)) \wedge \neg(\forall x(M(x) \rightarrow C(x)))$

1B. $\exists x, y (F(x) \wedge B(y) \wedge E(y, x))$

1C. $\forall x (M(x) \rightarrow E(x))$

2A. True. Take the following set $(x, y) = (x, 1)$. The condition $1 > x^2$ holds true $\forall x < 1$. By property of algebra, if $x < 1$, then $x^2 < x$. Therefore, it suffices that if $1 > x^2$, then $1 > x$, thereby proving that for a particular y , the condition holds true for every x provided the condition $y > x^2$ is met.

2B. False. Take the set $(x, y) = (x, -64)$. $\forall x$ such that $x = 2k + 1$, $\sqrt[x]{-64} < 0$. For sake of simplicity, let $x = 1$ and $x = 3$, respectively: $\sqrt[1]{-64} = -64 < 0$ and $\sqrt[3]{-64} = -4 < 0$. Therefore, it holds true $\sqrt[x]{y} < 0$ for multiple values of x , thereby disproving the statement.

2C. False. The condition $\sqrt[3]{x} \neq y^2$ can be simplified to $x \neq y^6$. Since $y^6 \geq 0 \forall y$, for the condition $x \neq y^6$ to hold true provided y can be any real number, x must be negative. However, if x is negative, the conclusion of the statement $x > 0$ does not stand. Therefore, the statement is false.

2D. False. Take the set $(x, y) = (0, 1)$. The set follows the condition $y > 0 \vee x < 0$. The set, however, fails to follow the condition of the statement's conclusion $x^y \geq y$ since $0 \not\geq 1$. Therefore, the statement with the domain $\forall x \forall y$ fails to be true for all x and for all y – as evidenced by the provided set.

2E. False. Take the set $(0.5, -0.5)$. The statement implies if one condition is true, then the other is, too. The statement $(0.5)^2 = (-0.5)^2$ holds true since $0.25 = 0.25$. However, the second statement $(0.5)^3 = (-0.5)^3$ does not hold true since $0.125 \neq -0.125$. Therefore, the statement with the domain $\forall x \forall y$ fails to be true for all x and for all y – as evidenced by the provided set.

3. $\exists! x P(x)$ can be written as: $(\exists x P(x)) \wedge (\forall x \forall y (P(x) \wedge P(y) \rightarrow x = y))$.

$$4. \exists x \forall y ((A(x) \rightarrow B(y)) \wedge (C(x, y) \rightarrow B(y))) \equiv \exists x \forall y (\neg B(y) \rightarrow (\neg A(x) \wedge \neg C(x, y)))$$

$\exists x \forall y ((A(x) \rightarrow B(y)) \wedge (C(x, y) \rightarrow B(y)))$	Given
$\exists x \forall y ((\neg A(x) \vee B(y)) \wedge (C(x, y) \rightarrow B(y)))$	Conjunctive Disjunction
$\exists x \forall y ((\neg A(x) \vee B(y)) \wedge (\neg C(x, y) \vee B(y)))$	Conjunctive Disjunction
$\exists x \forall y (B(y) \vee (\neg A(x) \wedge \neg C(x, y)))$	Distributive Laws
$\exists x \forall y (\neg B(y) \rightarrow (\neg A(x) \wedge \neg C(x, y)))$	Conjunctive Disjunction

$$5A. \forall x \neg \exists y \exists z (P(x, y) \leftrightarrow \neg S(y, z))$$

$\forall x \neg \exists y \exists z (P(x, y) \leftrightarrow \neg S(y, z))$	Given
$\forall x \forall y \forall z \neg (P(x, y) \leftrightarrow \neg S(y, z))$	Negation of Quantifiers
$\forall x \forall y \forall z \neg (((P(x, y) \rightarrow \neg S(y, z)) \wedge (\neg S(y, z) \rightarrow P(x, y)))$	Expansion of Biconditional Statement
$\forall x \forall y \forall z \neg (((\neg P(x, y) \vee \neg S(y, z)) \wedge (S(y, z) \vee P(x, y)))$	Conjunctive Disjunction
$\forall x \forall y \forall z ((P(x, y) \wedge S(y, z)) \vee (S(y, z) \wedge \neg P(x, y)))$	DeMorgan's Laws
$\forall x \forall y \forall z (S(y, z) \wedge (P(x, y) \vee \neg P(x, y)))$	Distributive Laws
$\forall x \forall y \forall z S(y, z) \wedge T$	Negation Laws
$\forall x \forall y \forall z S(y, z)$	Identity Laws

$$5B. \neg \forall y \neg \exists x (R(x, y) \rightarrow \exists z S(z, y))$$

$\neg \forall y \neg \exists x (R(x, y) \rightarrow \exists z S(z, y))$	Given
$\exists y \forall x \neg (R(x, y) \rightarrow \exists z S(z, y))$	Negation of Quantifiers
$\exists y \forall x \neg (\neg R(x, y) \vee \exists z S(z, y))$	Conjunctive Disjunction
$\exists y \forall x (R(x, y) \wedge \neg \exists z S(z, y))$	DeMorgan's Laws
$\exists y \forall x (R(x, y) \wedge \forall z \neg S(z, y))$	Negation of Quantifiers

6A. Take the set $(x, y) = (0, 0)$. The set meets the condition $y = x$, yet fails to assert the equality between $y^{\frac{1}{x}} \leq x^{-y}$ since $\frac{1}{0}$ is undefined. Therefore, the statement is false.

6B. By the quantifiers, the statement can be rewritten and described as for all y there exists a certain x and z such that $\sqrt[z]{x} = y$. If $z = 1$, then the 1^{th} root any number is that number due to properties of logarithms. Therefore, it is not possible to develop a counterexample.

6C. Take the set $(x, y) = (1, 1)$. The set meets the condition $y \leq x$ yet fails to meet either conclusion of $x^{17} > y$ or $y < 0$. Therefore, the statement is false.

7. Given a statement with two predicates, there exists an ambiguity when declaring the qualification of the variables before both predicates if both variables are the same. However, should two variables remain the same, then there exists no ambiguity since the qualification of said variable is directly before the respective predicate – as in the case of

the first statement. In the case of the second statement, two different variables are used and are assigned to a respective predicate, thereby removing ambiguity. Either method can be used and therefore is equivalent to the other statement.

8A. $\forall x (A(x) \rightarrow \neg C(x) \wedge \neg D(x) \wedge B(x))$

8B. $\neg \exists x (C(x) \wedge D(x) \wedge H(x)) \wedge \forall x (D(x))$

8C. $(\exists x (C(x) \wedge E(x)) \wedge \exists x (D(x) \wedge E(x)) \wedge \exists x (H(x) \wedge E(x)))$